

Algorithmic Trading and Market Quality: International Evidence

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Abstract

We study the effect of algorithmic trading (AT) on market quality between 2001 and 2011 in 42 equity markets around the world. We use exchange co-location service that increases AT as an exogenous instrument to draw causal inferences of AT on market quality. On average, AT improves liquidity and informational efficiency but increases short-term volatility. Importantly, AT also lowers execution shortfalls for buy-side institutional investors. Our results are surprisingly consistent across markets and thus across a wide range of AT environments. We further document that the beneficial effect of AT is stronger in large stocks than in small stocks.

JEL classification: G12, G14

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I. Introduction

By most accounts, high-frequency trading (HFT) represents the majority of the trading volume in today's equity markets.¹ Besides the sheer trading volume, HFT is important because the underlying strategies are often not transparent, nor are they well understood. These reasons elicit substantial public policy interest in the effects that HFT has on other market participants, trading strategies, and the quality of markets. Security market regulators around the world actively debate whether and, if so, how HFT should be regulated, and regularly scrutinize algorithmic and high-frequency order strategies and their consequences. Despite the intensity of this debate and a large theoretical and empirical literature in this area, many questions remain unanswered (see, e.g., Chordia et al. (2013), Jones (2013), O'Hara (2015), and Menkveld (2016) for reviews).

In this paper, we take a basic but comprehensive approach that contributes new insights and broad-sample evidence to this debate. Following Hendershott, Jones, and Menkveld (2011), we construct proxies for algorithmic trading (AT) intensity, a precondition for HFT, from order-related message traffic, and estimate an instrumental variable model that allows us to make causal inferences. We use eleven years of intraday data on security-level quotes and trades in 42 equity markets around the world, on average covering more than 21,507 firms per year. This new and comprehensive sample allows us to exploit the effect of AT on market quality across stocks, markets, and trading infrastructures.²

¹ HFT, also called low latency trading, refers to the activity of algorithms that emit orders or order cancellations, reacting within milli-(micro-)seconds to market updates or new information.

² Strictly speaking, HFT is a subset of AT. HFT likely accounts for most algorithmic message traffic and volume. Because our analysis of AT has implications for HFT, we use the terms "algorithmic trading" and "high-frequency trading" interchangeably.

In addition to examining the relationship between AT and standard market quality metrics of liquidity, informational efficiency, and short-term volatility, we investigate how AT impacts buy-side institutional investors. Market quality metrics are indicators of specific aspects of overall market quality. With HFT accounting for a large fraction of trading, market-wide statistics do not necessarily reflect the trading outcome for other market participants. For instance, institutional investors trade large quantities of shares and execute their orders over an extended period. This trading behavior increases the chances that these market participants become targets of predatory trading or “back running” strategies that some high-frequency traders use, hence leading to higher trading costs (Brunnermeier and Pedersen (2005), Yang and Zhu (2019)). Therefore, the net effect of higher AT intensity on buy-side institutional investors’ execution costs is an open empirical question. As an overall assessment of the impact of AT on institutional investors, we estimate and analyze execution shortfall, a commonly used trading cost measure applied to large institutional orders, using trade records of institutional investors around the world.

To help draw causal inferences of AT on market quality, we use the formal introduction of co-location services as an exogenous instrument for AT intensity, an instrument with the same interpretation across markets that differ in trading protocols and market structure. Co-location allows fast traders to minimize data turnaround time by physically locating their computer hardware close to the stock exchange’s hardware. The event itself, the actual introduction of co-location services, marks the exchange’s commitment to low-latency infrastructure. Other things equal, this commitment makes the stock exchange more attractive to AT and should thus lead to an increase in AT intensity. Conversely, the introduction of co-location services has no direct impact on market quality (Menkveld and Zoican (2017)). Therefore, we develop instruments based on co-location events and use them to help assess the effect of AT on market quality.

We find that AT affects various dimensions of market quality. First, higher AT intensity improves liquidity. Specifically, we find that higher AT intensity leads to narrower quoted and effective spreads. Second, more AT results in higher price efficiency. Stocks with higher AT intensity have a smaller absolute value of autocorrelations of intraday returns, an indicator of prices being more random. Third, higher AT intensity increases short-term volatility: daily price range, intraday return variances, and daily return volatility all increase with AT intensity.

We see at least two reasons why elevated volatility could be desirable. First, the more efficient markets are, the faster prices change in response to new information, producing presumably desirable price volatility. It is thus conceivable that the greater efficiency induced by more AT also produces higher desirable volatility. To address this issue empirically, we hold constant each stock's level of informational efficiency and still find that AT increases volatility. Therefore, it is unlikely that the AT-induced change in volatility is due solely to the "good" volatility associated with faster price discovery.

Second, volatility could increase when AT is more intense because some fast traders may prefer high-volatility environments. It is conceivable that these fast traders enter the market when volatility is high as a by-product of a market-making strategy. In this case, high volatility could be desirable because it attracts additional liquidity that would otherwise be absent. We analyze this possibility empirically by relating AT's effect on volatility to its contemporaneous effect on liquidity. We find that on days when AT leads to higher volatility, AT also induces lower liquidity. This link between volatility and liquidity suggests that either the same traders who generate higher volatility also cause lower liquidity by withdrawing their supply of liquidity or that the traders attracted by high volatility take liquidity rather than provide liquidity. Either way, AT activity that takes place during high-volatility episodes appears undesirable to the market.

Although we document higher AT-induced volatility, we are agnostic about the exact causes of such high volatility. More intense AT-induced volatility is consistent with the model

by Rosu (2019), who shows that volatility increases when more high-frequency traders enter the market or when their information becomes more precise. One might be concerned, however, that the release of firm fundamental news is driving the effect of AT on volatility. To mitigate this concern, we control for firm news announcements in our analysis. We find that volatility is indeed high at news announcements, but AT's effect on volatility extends well beyond the news event itself.

The multifaceted effects of AT on conventional measures of market quality motivate us to look at buy-side institutional transaction costs. For buy-side institutions, smaller spreads may help lower their trading costs, but higher information efficiency could increase their costs of trading with informed traders. Furthermore, higher volatility could increase their trading costs because large orders take longer time to execute and prices may be pushed away while their orders are being executed. Thus, ex ante it is not clear how execution shortfalls, which capture the cumulative price impact of a large order over the entire execution period rather than the half-spread for a single market order, are affected by AT. We find that the upside benefits of better liquidity are greater than the downside costs of increased volatility. The net effect of AT is beneficial to buy-side institutional investors: higher AT intensity reduces execution shortfalls.

While the average effect of AT on market quality has both upside benefits and downside costs, we observe substantial variation in AT intensity. Thus, understanding the cross-sectional differences in the benefits and costs of higher AT intensity is important. We assess how AT's effect on market quality varies with market capitalization. For example, providing liquidity in large-cap stocks typically requires less effort, and the trading intensity is high enough to allow for significant AT activity. Implementing market-making strategies or hiding informed trades is likely easier in large-cap stocks than in small-cap stocks. To analyze potential cross-sectional differences in the effects of AT, we divide sample stocks into three size groups based on market capitalization within each equity market and allow the impact of AT to vary across sizes.

We observe that the benefits of high AT intensity are more pronounced among large-cap stocks than in small-cap stocks. When AT increases, small stocks experience less liquidity improvement and little price efficiency enhancement. The main costs of high AT intensity in terms of elevated volatility are higher in small stocks than in large stocks. These findings indicate potential segmentation of liquidity provision across large and small stocks, suggesting that potential regulations on HFT should be cautious about whether one size fits all.

Our study contributes to the growing literature on HFT in the following ways. First, we provide new large-sample cross-country evidence of AT's effect on market quality. Existing research offers mixed evidence on the effect of HFT on market quality. Many studies find that HFT increases liquidity and price discovery. Still, others raise concerns about the quality or usefulness of HFT-provided liquidity, noting that such liquidity is often short-lived and negatively affects information acquisition and volatility.³ This is partly due to the fact that most existing studies either employ a small sample of U.S. stocks traded by HFT firms or rely on a particular market outside the U.S.⁴ Our analysis of a comprehensive international sample covering a period when AT starts to mature around the globe helps draw a broader inference on the effect of AT on market quality.

Second, our analysis of the effect of AT on volatility sheds light on a relatively overlooked issue that has potential policy implications. In contrast to prior research that finds lower volatility from market-making HFT based on small samples (e.g., Hagströmer and Nordén (2013)), our study shows that across a large number of equity markets, more intense AT activity

³ For example, among others, Brogaard, Hendershott, and Riordan (2014) find improved liquidity by HFT. Hasbrouck and Saar (2009) document the “fleeting” nature of many limit orders in electronic markets, questioning the traditional view that limit orders provide liquidity to the market. Weller (2018) finds that AT can reduce price informativeness due to less information acquisition.

⁴ For example, among studies that look at markets outside the U.S., Hendershott and Riordan (2013) examine 30 German stocks in a stock index on the Deutsche Boerse in January 2008; Menkveld (2013) examines one HFT firm on Euronext and Chi-X; Brogaard, Hagstroemer, Norden, and Riordan (2015) examine NASDAQ OMX Stockholm; Anand and Venkataraman (2016) look at Toronto Stock Exchange; van Kervel and Menkveld (2019) examine Swedish market for large-cap stocks.

induces higher volatility. Our finding contributes to the theoretical debate on the relation between high-frequency market-making and volatility (see, e.g., Baruch and Glosten (2016), Yueshen (2017), and Rosu (2019)).

Third, our study goes beyond examining AT's effect on traditional trading cost measures such as bid-ask spreads. Our analysis of buy-side institutional investors improves our understanding of how AT affects this important group of market participants. We show that these institutions incur lower execution costs when AT intensity is higher. Our international evidence complements recent studies that examine execution costs on a specific market (e.g., van Kervel and Menkveld (2019) on Swedish index stocks; Korajczyk and Murphy (2019) on the Canadian market; and Tong (2015) on Nasdaq).

Fourth, we establish differences across stock sizes in the effect of AT on market quality. This is important, because given the increasing focus on how to regulate HFT, our results suggest segmentation of liquidity provision and imply that optimal regulation may need to impose different rules on different categories of stocks.

The rest of the paper is organized as follows. In Section II, we discuss our data sources and define key variables used in our analysis. We discuss our empirical design in Section III and present our main results in Section IV. Section V provides additional analysis and discussion. Section VI concludes.

II. Data and Variables

A. Data Sources

We combine several data sources in our analysis. The Thomson Reuters Tick History (TRTH) database contains intraday trades and quotes for many equity markets around the world. Intraday trades and quotes for U.S. stocks listed on NYSE and NASDAQ are from the Trades and Quotes (TAQ) database. We use Thomson Reuters Datastream and Center for Research and

Security Prices (CRSP) to obtain stock-level and market-level data. In addition, data on buy-side institutional transaction costs come from the Ancerno database compiled by Ancerno Ltd. (formerly the Abel/Noser Corporation). Finally, we use RavenPack to obtain data on a company's news release.

The TRTH database (supplied by the Securities Industry Research Centre of Asia-Pacific, SIRCA) provides access to the data feeds from various stock and derivatives exchanges that are time-stamped to the millisecond and transmitted through Reuters' terminals. TRTH organizes data by the Reuters Instrument Code (RIC). Each RIC is associated with a list of characteristics, such as asset class (e.g., equity), market, currency denomination, the date of the first and the last record, and the ISIN and SEDOL where applicable. The database contains more than 5 million equity and derivatives instruments around the world. A company may have multiple RICs representing common shares, preferred shares, different share classes, or securities in special trading status. To both create a comprehensive sample of RICs for each market and avoid double-counting, we focus on one common stock per company, traded in the home country and in local currency. As TRTH has limited coverage of these screening variables, we construct our sample by first merging TRTH with Datastream by identifying matches between RIC and Datastream firm identifiers.

Datastream identifies securities by DSCODE, which uniquely identifies a security-trading venue combination. Each DSCODE is associated with a comprehensive list of static securities information. We retain only the DSCODE in the local market, traded in the local currency and identified as "major security" and "primary quote." These screening criteria lead to one DSCODE per domestic company, each having a unique ISIN. We are interested in the primary trading location, which coincides with the listing exchange in all markets except Germany. For Germany, we use XETRA (the primary trading location) rather than Frankfurt (the primary listing location), because XETRA handles roughly 90% of volume for most stocks in our sample

period. We merge the two data sources as follows: For each exchange, we obtain the ISIN and the history of high, low, and last trade price for each RIC from the TRTH database. We find the corresponding trading venue on Datastream and identify the unadjusted daily price, market capitalization, and the adjustment factor (dilution) for each screened DSCODE. Then we match RIC to DSCODE using ISIN. There may be more than one RIC per DSCODE if a company changes the trading symbol. We validate the match by comparing the Datastream price history to the TRTH price history after adjusting for currency-reporting differences.⁵ This procedure, together with TAQ, produces a sample of stocks trading on 42 equity exchanges in 37 countries spanning from 2001 to 2011.⁶

The TRTH data have qualifiers that contain market-specific codes denoting the first trade of the day, auction trades, and irregular trades (such as off-market trades or option exercises). We remove irregular trades before computing intraday variables.

Trading hours differ across exchanges and over time. We determine each exchange's historical trading hour regime by examining the trade frequency across all stocks on the exchange at 5-minute intervals. We identify the opening and closing times of regular trading from spikes and drops in trading activity across all stocks at each exchange. We cross-check our approach against the trading hour regime and the trading mechanism entries listed in Reuters' Speedguide and the Handbook of World Stock, Derivative and Commodity Exchanges.

We also employ data from Ancerno in our analysis. Ancerno provides transaction costs analysis for its institutional buy-side clients. Each record in Ancerno database includes an

⁵ TRTH prices are historical prices in the original currency. Datastream unadjusted prices are historical prices in the current currency unit, e.g., French stocks prior to 1999 were traded in French franc. We convert Datastream prices to Euro equivalents.

⁶ We drop Ireland, where data is available for fewer than 30 stocks prior to 2008. China has three exchanges covered in Datastream (Hong Kong, Shenzhen, and Shanghai); India (Mumbai and National exchanges), Japan (Tokyo and Osaka), and the U.S. (NYSE and Nasdaq) have two; and all other countries have one exchange included in the sample.

anonymized client code, a broker code, the CUSIP and ISIN for the stock, the date of execution, the execution price, and the number of shares executed, as well as whether the execution is a buy or sell. Multiple trades are often associated with a client on a particular stock day. We match the Ancerno data to CRSP and Datastream data using date, CUSIP, ISIN, and ticker. To accommodate investors who split orders across brokers, we follow Anand, Irvine, Puckett, and Venkataraman (2012) by aggregating trades into daily orders by client, stock, date, and trade direction.

RavenPack, a leading news provider that covers corporate news releases, is used for some of our analysis. RavenPack generates company relevance scores ranging from 0 to 100 to measure the informational content of a news story. The company relevance scores allow researchers to extract news articles that are related to a specific company in the database. A relevance score equal to 100 indicates that a company is quoted as the main subject of a news release. Following Dai, Parwada, and Zhang (2015), we record the presence of news articles that are related to a specific company with relevance scores equal to 100 on a particular day.

B. Variable Construction

Following previous literature, we construct variables that describe several dimensions of market quality, focusing on liquidity, informational efficiency, and volatility. We also construct measures for buy-side institutional trading costs. We describe these variables in this subsection, along with our proxy for AT.

1. Liquidity

We compute two standard measures of liquidity. For each stock, we have the best quoted spreads throughout the trading day. For a given time interval s , the relative quoted spread, RQS, is defined as

$$(1) \quad RQS_s = (ASK_s - BID_s) / ((ASK_s + BID_s)/2)$$

where ASK_s is the best ask quote and BID_s is the best bid quote in that time interval. We standardize the quoted bid-ask spread by the quote midpoint. When aggregating over a trading day, we use the time-weighted average of RQS_s . Narrower RQS suggests better liquidity.

To take into account possible price improvement arising from hidden liquidity, we compute the relative effective spread, RES. RES on the k^{th} trade is defined as

$$(2) \quad RES_k = 2D_k (PRICE_k - MP_k) / MP_k$$

where D_k is an indicator variable that equals +1 if the k^{th} trade is buyer initiated and -1 if it is seller initiated, $PRICE_k$ is the price of the k^{th} trade, and MP_k is the prevailing midpoint at the time of the trade. We follow the standard trade-signing approach of Lee and Ready (1991) and use contemporaneous quotes to sign trades and calculate effective spreads (Bessembinder (2003)). We then calculate the trade-weighted average across all trades that day for each stock. RES measures price changes associated with trading. The narrower the relative effective spread, the more liquid is the stock.

2. Informational Efficiency

We compute intraday measures of relative informational efficiency following Boehmer and Kelley (2009). Specifically, we rely on intraday measures of quote midpoint autocorrelation. If prices are efficient and follow a random walk, these measures should be close to zero. Deviations from zero in either a positive or negative direction indicate partial predictability and less efficient prices. We thus use the absolute value of quote midpoint return autocorrelations. The larger this value is, the less efficient the prices are. We estimate this measure for each stock-day, $|AR10|$ ($|AR30|$), based on 10- (30-) minute return intervals (Chordia, Roll, and Subrahmanyam (2005)).

3. Volatility

Our primary measure of a stock's volatility is PRICE_RANGE defined as the intraday range between the highest and lowest prices of a day standardized by the daily closing price.

This measure is a reasonable proxy because it reflects intraday fluctuations in share prices that may trigger or result from algorithmic trading. As higher-frequency measures, we also use the log of intra-day return variances computed from 10-minute and 30-minute mid-quote returns, $\ln(\text{RET10_VAR})$ and $\ln(\text{RET30_VAR})$. Additionally, we calculate realized volatility using the absolute value of daily returns ($|\text{RET}|$).

4. Execution Shortfall

In addition to the above traditional measures, we estimate execution shortfall for the buy-side institutions that report to Ancerno following Anand et al. (2012). In contrast to the trade-and-quote-based measures, the shortfall represents actual execution expenses for an institution's order flow. We define execution shortfall, SHORTFALL , as:

$$(3) \quad \text{SHORTFALL} = D (XP - RP) / RP$$

where D is an indicator variable that equals +1 if the order is a buy and -1 if it is a sell, XP is the volume-weighted average price across component trades of a daily order, and RP is the reference price, defined as the opening price on the day of the order.

5. Proxy for AT

The growth of AT activity since the early 2000s has spurred many studies to develop measures of AT and examine its consequences. Some studies are able to observe the actual identifier for a group of high-frequency traders. For example, within the U.S. market, Brogaard et al. (2014), Brogaard et al. (2015), and Carrion (2013) use a 2008-2009 Nasdaq sample that summarizes the aggregate order flow generated by 26 HFT firms that capture about three quarters of trading volume in the sample stocks. A few other studies have access to actual HFT

data for a specific foreign market.⁷ Most of the remaining studies (see, e.g., Hendershott, Jones, and Menkveld (2011), Hasbrouck and Saar (2013)) use some variations of message-to-trade ratios as proxies for AT/HFT. Messages refer to trades, order arrivals, or order cancellations. Because many AT strategies involve frequent cancel-and-replace order traffic, the proportion of traffic that leads to a trade is typically much smaller for algorithmic traders than for non-algorithmic traders. Message-to-trade ratios are well accepted as proxies for AT activity in the trading industry. They allow researchers to use the full panel of stock-days for which standard intraday trade and quote data are available and, in principle, allow the strongest inferences.

We follow Hendershott et al. (2011) and construct a proxy for AT that reflects fast order submissions and cancellations that are associated with algorithmic trading strategies. Specifically, our proxy for AT is the negative of trading volume in USD100 divided by the number of messages. It represents the negative of the dollar volume associated with each message (defined as either a trade or a quote update). An increase in this measure reflects an increase in AT activity.

Our AT measure is particularly well suited for international and inter-firm comparisons, because it provides a continuous scale of *relative* AT intensity for each market (rather than an on-off switch, or an absolute measure that does not recognize differences across markets). It thus allows us to use the same measure across a variety of market structures that differ substantially in the degree to which AT is prevalent. Perhaps more importantly, using a relative measure of AT intensity allows the nature of “fast” or “low-latency” trading to differ across markets. For example, some markets impose hurdles to fast quoting. AT will remain more intense in some

⁷ For example, outside the U.S., Hendershott and Riordan (2013) examine 30 Deutscher Aktien Index stocks on the Deutsche Boerse in January 2008. Menkveld (2013) examines one HFT firm on Euronext and Chi-X. Brogaard et al. (2015) find NASDAQ OMX Stockholm introduction of optional colocation improve effective spread and depth for slow traders and the market overall. Anand and Venkataraman (2016) find that synchronous withdrawal of liquidity provision by HFT firms under difficult market conditions contributes to fragility of liquidity supply on Toronto Stock Exchange.

stocks and some episodes than in others traded on the same exchange. Moreover, because our proxy represents a relative measure of AT, we can use it to compare the effect of AT across countries even when comparing a market with latency measured in nanoseconds to one where it is measured in seconds. In either market, high-frequency traders gain by being faster than other traders and our relative, continuous measure of AT can capture this contrast well.⁸

Our measure of AT differs in an important way from the one used by Hendershott, Jones, and Menkveld (2011), who have access to order-level messages. For our worldwide sample, we have access only to a subset of these messages, observing each exchange's best quotes and trades, rather than all order-related messages. Conceptually, using just trades and changes in the best quotes should not impose serious problems on our analysis. For example, the HFT strategies mentioned in the SEC 2010 concept release involve most activity *at* the BBO, rather than *behind* it. Therefore, the AT activity in our BBO trade dataset is highly correlated with AT activity in an order trade dataset.⁹

⁸ The AT proxy also has potential drawbacks. First, as suggested by Hendershott et al. (2011), it may have low power for small stocks due to less algorithmic trading. For small stocks the AT proxy could potentially capture market uncertainty, when orders are submitted and cancelled yet few of those orders result in transactions. To mitigate this concern, we explicitly control for market cap in our regression analysis. Second, the AT proxy may not be very good during the release of news announcements. Fundamental volatility is high during news announcements, and the event, rather than algorithmic trading, could cause a high number of electronic messages to transaction volume ratio and high price volatility. To help address this concern, we control for news announcements in additional analysis in Section IV.C. Our results suggest that the AT measure is a reasonable proxy for algorithmic trading even in the presence of high fundamental volatility during news releases.

⁹ We formally address the correlation between AT measures based on order level data, and AT based on trades and best quotes. We repeat Hendershott et al. (2011)'s time-series and panel results for the U.S. using order-level data and compare the results to the ones we obtain with our data and our version of the AT measure. The time series in which our order level data and the TAQ data overlap is very similar to the period presented in Hendershott et al. (2011). Our exercise using NYSE activity yields qualitatively identical results for their order-level count and our count of trades and inside quote changes. This result is not surprising because the correlation of these two series, for the average stock, exceeds 0.9. Therefore, we have little reason to expect our AT proxy to deliver substantially different results than the proxy in Hendershott et al. (2011).

III. Sample and Methodology

A. Descriptive Statistics

To be included in our analysis, a stock needs to have data for more than 21 trading days during the sample period to minimize the concern of thin trading. We then winsorize all variables each day at 0.5% and 99.5% within each market. To illustrate the breadth of our sample, Table 1 lists the number of stocks for each market. For the average year, our sample includes about 21,507 firms, and we have substantial variation across markets. Over the sample period, the number of listed firms increases, on average, by more than half, or from 473 to 560.

Our key variable of interest is the distribution of message traffic (the number of all quote changes and all trades), the main component of our proxy for algorithmic trading activity. For each market, Table 2 lists the average number of messages per stock day in 2001 and in 2011 along with the growth over this eleven-year period. We make several important observations. First, message traffic grows over time, often many folds, except for Greece. Overall, message traffic grows by more than ten times across markets, from 289 messages per stock day in the beginning of the sample period to 4,042 in 2011. This development is consistent with AT playing an increasingly important role in trading around the world. Figure 1 shows that this growth accelerates exponentially during the second half of the decade when exchanges started to offer co-location services. Figure 1 also reveals that most of the message growth comes from quote messages rather than trade messages, further motivating our proxy for the unobservable AT activity.¹⁰

Figure 2 presents the monthly time-series of liquidity, efficiency, and volatility, respectively. For each market day, we first compute an equally-weighted average across stocks,

¹⁰ Conrad, Wahal and Xiang (2015) make similar observations that HFT mainly act via quotes than trades for the U.S. and Japan markets.

and then calculate the average within each market month. In the figure, we plot the monthly time series of averages across markets. The relative quoted and effective spreads, RQS and RES, in Panel A show similar patterns. For example, RES began at 250 basis points (bps) in the beginning of 2001 and declined to 150 bps by the end of 2007. Afterwards, it peaked at the end of 2008, when the financial crises around the globe started to unfold. RQS declined from 500 bps in 2001 to 246 bps in 2007, and then rose to 660 bps during the financial crisis before declining to 400 bps in 2011. The difference between RQS and RES primarily reflects the absence of trades during high-spread periods or the presence of traders who execute against non-displayed liquidity inside the quotes.

Panel B plots two efficiency measures, $|AR10|$ and $|AR30|$. Both measures show a slight decrease over the sample period. The intraday volatility measures in Panel C decline slightly over the first half of the sample period, with a large spike towards the end of 2008 and a smaller increase in 2011.

Panel D graphs the monthly measures of execution shortfall. We see an increase in execution shortfall from below 20 bps in the pre-crisis period to more than 35 bps during the financial crisis period followed by a decline in the post-crisis period.

B. Methodology

To establish a causal relation between AT and market quality, we use an instrumental variable approach. We seek an instrument that satisfies the exclusion restriction, i.e., it is not causally related to any of our market quality variables. In addition, the instrument should be closely related to AT intensity. As our sample represents a multitude of trading protocols and market structures, finding an instrument that has the same interpretation across markets is important. We rely on the event of co-location in each stock market. Co-location refers to locating a trader's computer hardware physically close to a trading center's hardware. Doing so allows the trader's order submission algorithm to interact with the trading center with minimal

latency. Brogaard et al. (2015) show that co-location (at NASDAQ OMX Stockholm) allows fast traders to reduce their cost of liquidity provision and thus trade more profitably. Similarly, Baron et al. (2019) find that HFT firms that improve their latency rank due to co-location upgrades (at NASDAQ OMX Stockholm) deliver better trading performance that comes through short-lived information channels as well as risk management channels.¹¹

To introduce a co-location service, some markets announce a program or pricing scheme, while others announce that a specific trading firm is co-located (and typically invite successors). From these announcements, we identify the first implementation date (rather than use the first announcement date) to capture the change in trading that is prompted by the lower co-location-related latency. Co-location introductions mark events that are fairly homogenous across exchanges, in that the event specifically provides the infrastructure for fast traders and signals an exchange's commitment to accommodating such traders. As a potential caveat, the precision of reporting on co-location event dates could differ across countries. Yet, to the extent that the resulting errors are random, they should not affect the consistency of the IV estimator because such random errors would be captured by the regression error.

During our sample period, 22 markets have adopted co-location services (see the list of co-location dates in Appendix 1).¹² To allow a reasonable amount of time to observe any potential changes in the effects of AT on market quality due to co-location, we select a two-year window centered on the first co-location implementation. Specifically, the indicator variable `CO_LOCATION` takes a value of zero in the 12 months leading to the adoption of co-location

¹¹ Other possible instruments include the introduction of direct market access for traders, DMA, or other updates to the trading protocol that imply a structural change in how traders implement AT/HFT strategies. We believe that co-location allows the cleanest measurement. Moreover, the first co-location service offered in a market also tends to be prominently and consistently reported by the media.

¹² For two markets (Germany and Taiwan) we couldn't obtain the specific month during which co-location was implemented, so we use the year-end as the event month. NYSE's co-location was first offered through its acquisition of TransactTools in January 2007 and its most recent co-location data service started in August 2010.

and switches to one during the 12 months after co-location. To account for the clustering of co-location events across markets, robust standard errors are clustered by date.

To implement our instrumental variable approach, we regress our AT proxy on CO_LOCATION in the first stage to obtain the predicted value of AT.¹³ The indicator variable CO_LOCATION has a coefficient of 0.108 (*t-stat* = 13.1). This suggests that the adoption of co-location services significantly contributes to AT growth.

In the second stage, we estimate the following specification:

$$(4) \quad MQ_{it} = \alpha + \beta AT^*_{it} + \delta X_{it} + \varepsilon_{it}$$

Where *i*, *t* refer to stock and day, respectively, *AT*^{*} is the predicted value from the first-stage regression, and *X* is a vector of control variables. Control variables include daily share turnover, TURN_OVER, the inverse of share price, INVERSE_PRC, the natural log of market capitalization, ln(MKT_CAP), PRICE_RANGE (omitted from the volatility regressions which add RES and |AR30|), and the first lag of the dependent variable. These control variables have been shown to be related to market quality (e.g., Hendershott et al. (2011)). Following Boehmer and Kelley (2009), we lag all control variables by one day to ensure that explanatory variables are predetermined.¹⁴ To make comparisons meaningful across firms and markets, all continuous variables are standardized to have a zero mean and unit variance within each firm during the co-location event window.

IV. Empirical Results

¹³ Note that in the first stage regression, we regress AT only on CO_LOCATION. This approach differs from a conventional IV approach where only one dependent variable is examined and the same set of control variables is used in both the first and the second stage regressions. Since our control variables include the lagged value of market quality measure, using the conventional IV approach would yield a different coefficient on the indicator variable CO_LOCATION and hence a different predicted value of AT used in the second stage regression for each measure of market quality.

¹⁴ Using contemporaneous values produces similar results.

In this section, we first analyze how AT affects measures of liquidity, price efficiency, short-term volatility, and institutional buy-side execution costs. We then examine whether there exist any cross-sectional variations in the AT effect on these dimensions of market quality.

A. Main Results

Table 3 presents the second-stage regression estimation results based on equation (4). Panel A reports the effect of AT on liquidity measures of RQS and RES after controlling for other variables.¹⁵ We find that AT significantly lowers both the quoted and effective spreads. For example, a one standard deviation increase in AT intensity improves the relative effective spreads by 0.7 standard deviations.

The beneficial effect of AT on liquidity we document across the international markets is in line with several existing studies. For example, Hendershott et al. (2011) are among the first to show that algorithmic trading leads to better liquidity. They use the 2003 introduction of autoquote at the NYSE as an instrument to establish causality from algorithmic trading to market quality improvements. Using HFT activity inferred from millisecond-level responses, Hasbrouck and Saar (2013) find improvements in spreads when fast traders are more active. Malinova, Park, and Riordan (2013) show that a decline in HFT reduces liquidity for retail traders in the Canadian market.

Panel B presents the effect of AT on informational efficiency. The significantly negative coefficients on AT* suggest that more intense AT activity leads to a smaller absolute value of intraday return autocorrelations, indicating that the movement of prices is closer to a random walk. These results indicate that across markets, more AT consistently results in improvements in informational efficiency. This finding complements Brogaard et al. (2014) who show that HFT facilitates price discovery among a sample of Nasdaq stocks. Conrad et al. (2015) also find

¹⁵ Coefficients on control variables are not tabulated.

higher quoting activity proxying for higher HFT activity is associated with better price efficiency.

Panel C presents the effect of AT on short-term volatility. We find that more intense AT leads to higher volatility, and the results are uniform across volatility proxies whether we look at intraday realized volatility (PRICE_RANGE , $\ln(\text{RET10_VAR})$, $\ln(\text{RET30_VAR})$) or daily realized volatility ($|\text{RET}|$). Larger AT-induced short-term volatility is also documented in several other studies in different contexts. For example, Kirilenko, Kyle, Samadi, and Tuzun (2017) argue that HFT worsened (but did not cause) the May 6, 2010 flash crash. Dichev, Huang, and Zhou (2014) show that trading per se generates excess volatility, arguing that HFT can lead to undesirable levels of volatility.

Volatility is important to traders and can have adverse effects on market quality through several channels. Limit orders provide liquidity to the market and represent options to trade for other market participants. Greater volatility makes this option more expensive and thus makes liquidity provision more costly.¹⁶ We will return to the volatility discussion in Section IV.C.

In Panel D, we examine the impact of AT intensity on execution shortfall, an overall assessment of the execution costs incurred by buy-side institutional investors. As alluded to earlier, market-wide metrics of trading costs may not properly represent the costs to buy-side institutional traders, because they typically trade large quantities of shares and their orders are executed over a longer trading horizon. High-frequency traders can choose to provide liquidity or conduct order-anticipation strategies and back-running strategies on large institutional orders

¹⁶ Consistent with this concern, Egginton, Van Ness, and Van Ness (2016) show that periods of extremely active quoting behaviour are associated with degraded liquidity and elevated volatility. Importantly, they show that such episodes are surprisingly frequent. Yet despite good economic reasons for such quote-bunching to occur as a benign by-product of HF liquidity provision, as Hasbrouck and Saar (2013) argue, that this quote-bunching arises as a consequence of intentional “quote stuffing” is also possible. This practice involves submitting a large volume of messages to disguise trading strategies. Gai, Yao, and Ye (2014) show that quote stuffing has negative effects on trading.

(van Kervel and Menkveld (2019), Yang and Zhu (2019)). While narrower spreads can lower trading costs for an average trader who submits a market order, higher information efficiency may increase the costs of trading with informed traders. Moreover, higher volatility could lead to higher trading cost variability and increase execution risk. Thus, the net effect of higher AT intensity on institutional investors' execution costs is best answered by the data.¹⁷

Panel D reports the effect of AT on execution cost, SHORTFALL. We find a significant and negative coefficient on AT*. This suggests that more AT helps reduce buy-side execution shortfall. Thus, the net effects of smaller spreads, better information efficiency and higher volatility induced by higher AT intensity appear beneficial to this group of market participants. Our finding based on a broad international sample helps resolve mixed results reported in recent studies that examine institutional investors' execution costs on a specific market (e.g., Tong (2015), van Kervel and Menkveld (2019), and Korajczyk and Murphy (2019)).

B. Heterogeneity of AT Impact

Both anecdotal and academic evidence suggests that AT can be more important in larger stocks than in smaller stocks. We examine potential heterogeneity in the impact of AT on market quality across different market-cap groups. We group stocks into small, medium, and large categories based on their market capitalization within each market. Then we interact the instrumented AT with three size indicator variables in the second stage to capture different AT

¹⁷ O'Hara (2015) proposes that with the increasing presence of HFT, the rest of the market, including large institutional investors, becomes more sophisticated by adapting trading strategies to protect their interests from high frequency traders. For example, if market makers and other traders split orders more following the co-location events, institutional investors are expected to adjust their trading algorithm to such changes. Otherwise, orders of institutional investors might risk standing out. We compute an institutional client's order split ratios by dividing the number of trades across client day orders by the number of client day orders. We compare this ratio in the pre-colocation period to that in the post-colocation period, and find an increase in splitting orders by institutional investors. These results suggest that institutional investors appear to adapt their order splitting behavior to the changing trading environment, consistent with the feature of market ecology proposed in O'Hara (2015).

impact across market-capitalizations. Specifically, we expand equation (4) to the following specification:

$$(5) \quad MQ_{it} = \alpha + \beta_1 AT_{it}^* \times S_i + \beta_2 AT_{it}^* \times M_i + \beta_3 AT_{it}^* \times L_i + \delta X_{it} + \varepsilon_{it}$$

where S , M , L represent indicator variables for small, medium, and large stocks, respectively.¹⁸

All other variables are defined in equation (4).

Table 4 presents the estimation results. Panel A focuses on liquidity measures. We find that AT improves liquidity in all size categories, but its effect is stronger in large stocks. Take RES for example, the negative coefficient on $AT^* \times L$ is almost twice as large as the coefficient on $AT^* \times S$, suggesting a much larger reduction in effective spreads in large-cap stocks compared to small-cap stocks. Panel B reports AT's effect on price efficiency across three sizes. Again, AT's effect is mainly present in medium and large stocks. AT has minimal improvement in price efficiency in small-cap stocks. Panel C presents the volatility results. Although we see higher volatility across all sizes, smaller stocks experience a larger increase in volatility (except for the daily price range, which is more sensitive for large stocks).

Lastly, Panel D shows that the beneficial effect of AT on execution shortfalls is mostly concentrated in large and medium stocks. Note that the negative (but insignificant) coefficient on the interaction between AT and small stocks indicates that AT does not lead to lower execution cost when buy-side institutional investors trade small stocks.

Overall, results in this subsection suggest heterogeneity in AT's effect on market quality. AT leads to better liquidity, faster price efficiency, moderately higher volatility and lower execution costs in large stocks. Small stocks, however, experience less liquidity improvement and much higher volatility. These results indicate that potential segmentation in liquidity

¹⁸ Regressions also include the size indicator variables but are not tabulated.

provision across stocks of different sizes goes beyond the U.S. market (Hendershott et al. (2011)).

C. AT and Volatility

We show earlier that higher AT intensity leads to higher volatility. This finding raises questions regarding the nature of this volatility. Given that more AT leads to better informational efficiency, it is conceivable that the elevated volatility reflects faster price adjustments when new information arrives. Under this scenario, the higher volatility reflects new information, not noise, and could therefore be desirable. Another possibility is that narrower spreads, which also result from greater AT activity, are related to smaller quoted sizes, so that subsequent trades result in trade prints that experience lower trade-by-trade execution costs but generate greater price fluctuations. Such a trade-off between liquidity and volatility could be desirable if the benefit of smaller spreads outweighs the potential cost of elevated trade-to-trade volatility. We take several approaches to address these possibilities.

First, we directly control for price efficiency and liquidity in the volatility regression. Specifically, efficiency proxy $|AR30|$ and liquidity proxy RES are already included in the regression model whenever we estimate AT on volatility in equation (4). By controlling for efficiency and round-trip transaction costs, we hold constant price efficiency and liquidity which likely are the main sources of “good” volatility. Since the reported results already control for informational efficiency and liquidity associated with “good” volatility, it is difficult to attribute the elevated volatility that we observe to either faster incorporation of new information or to tighter spreads.

Second, we examine whether higher AT-induced volatility is associated with improved liquidity. We show that more AT leads to better liquidity and greater efficiency but also greater volatility. If a stock experiences these effects on the same trading day, they could conceivably offset one another. For example, high-volatility periods could attract AT, which would then lead

to liquidity improvement. Although we cannot fully disentangle these effects and causal directions, we conduct a simple test that sheds additional light on the relation among AT, volatility, and liquidity. We employ a two-step procedure. In the first step, we estimate cross-sectional regression within each market day, regressing liquidity, price efficiency, and volatility on AT and control variables, and record the AT coefficients. Doing so produces a time series of daily AT coefficients for each market; one set each for liquidity, efficiency, and volatility. In the second step, we compute Spearman rank correlations between liquidity and volatility effects and between efficiency and volatility effects.¹⁹

Panel A of Table 5 reports Spearman rank correlations between AT coefficients for liquidity and volatility. All but one of the correlations is positive, and most are significantly so. This means that on days when AT is associated with higher volatility, AT is also contemporaneously related to wider spreads, rather than narrower spreads. Or, conversely, if high volatility indeed attracts algorithmic traders, these traders demand—rather than supply—liquidity. Therefore, at least in our sample, the costs of high volatility are not contemporaneously offset by greater liquidity, as suggested by Castura, Litzenberger, Gorelick, and Dwivedi (2010).

In contrast to the relationship between liquidity and volatility, Panel B shows that days with high efficiency also tend to have high volatility. This result is intuitive, because greater efficiency implies faster incorporation of news into prices, resulting in greater realized volatility. Because our volatility regressions already control for the level of efficiency, this observation does not affect our inferences from Panel A. In other words, the greater volatility-related efficiency happens on days when liquidity declines.

Third, we address a potential concern that fundamental volatility tends to be high during news announcements, and the news event, rather than algorithmic trading, causes higher

¹⁹ Pearson correlations produce the same inferences.

volatility. To examine this possibility, we explicitly control for news announcements in our regression. Specifically, we obtain news data from RavenPack and augment equation (4) into the following specification:

$$(6) \quad MQ_{it} = \alpha + \beta_1 AT_{it}^* + \beta_2 AT_{it}^* \times NEWS_{it} + \beta_3 NEWS_{it} + \delta X_{it} + \varepsilon_{it},$$

where NEWS is an indicator variable equal to one if there is news announcement for a firm on a day, and zero otherwise (Dai et al. (2015)).

Table 6 reports the regression results from estimating equation (6). As expected, the significant coefficient on NEWS suggests that volatility is heightened when the news about a firm is made public. More importantly, the coefficient on AT^* remains significant and positive, suggesting that more AT leads to higher volatility even in the absence of news. The interaction term of $AT^* \times NEWS$ indicates that AT's effect on volatility is even stronger on news days.

Overall, these three approaches complement one another in addressing the nature of the volatility. Although we are agnostic about the exact causes of AT-induced volatility, the overall evidence suggests that AT-induced volatility is difficult to be attributed to “good” volatility that would arise from faster price discovery, to algorithmic traders' inclination to enter the market when volatility is high, or to trading on fundamental news. Although we see beneficial effects of higher AT activity in improving liquidity and efficiency, more AT also appears to elevate undesirable volatility.

D. AT and Liquidity

We have shown that AT affects the nature of trading and improves liquidity by narrowing spreads for market participants. A natural question is how AT affects the permanent and transitory price impact patterns of trades, two components of relative effective spreads. Following the literature, we decompose RES into relative price impact (RPI) and relative realized spreads (RRS). RPI captures gross loss due to informed liquidity demanders, and RRS represents compensation to market makers for providing liquidity.

Similar to Conrad and Wahal (2020), we decompose RES into RPI and RRS over various horizons shorter than the conventional five-minute window which may no longer be appropriate in the high-frequency trading environment. Different from Conrad and Wahal (2020) who study the U.S. equity market in a more recent period which offers enough quoting and trading activities at higher frequencies, stock markets in our international sample vary in their market structure and quoting/trading intensity during 2001-2011. Specifically, the number of quotes per minute for a typical stock among the international markets averages at 3.3 in our sample, or less than 4 quote updates within a minute. This suggests that it is more meaningful to analyze horizons over 30 seconds in our sample. We thus estimate RPI and RRS using horizons of 30, 60, 90, 120 seconds following a trade. Specifically, RPI on the k^{th} trade is defined as

$$(7) \quad \text{RPI}_{-t} = 2D_k (\text{MP}_{k+t} - \text{MP}_k) / \text{MP}_k$$

where D_k is an indicator variable that equals +1 (-1) if the k^{th} trade is a buyer (seller) initiated, and MP_{k+t} is the midpoint t seconds after the trade, with t taking the value of 30, 60, 90, and 120, respectively. RPI measures the change in quote midpoints that is attributable to the information content of the trade.

RRS on the k^{th} trade is defined as

$$(8) \quad \text{RRS}_{-t} = 2D_k (\text{PRICE}_k - \text{MP}_{k+t}) / \text{MP}_k$$

For regression analysis, we compute trade-weighted averages of RPI and RRS for each stock-day. Specifically, we regress $\text{RPI}_{-30(60,90,120)}$ and $\text{RRS}_{-30(60,90,120)}$ on the instrumented AT and controls as specified in equation (4).

We report the coefficients on AT* in Table 7. Panel A has RPI of various horizons as the dependent variable. We find that higher AT intensity reduces $\text{RPI}_{-30(60,90,120)}$, suggesting that the decline in adverse selection losses to liquidity demanders contributes to narrowed relative effective spreads. In addition, the magnitude of the coefficient on AT* is largest at the 30-second horizon and decreases monotonically with the trading horizon, indicating that price impact

occurs more quickly as AT increases. Panel B has RRS of various horizons as the dependent variable. We find the coefficient on AT* is negative and statistically significant at 10% for RRS₃₀, and is insignificant as trading horizons become longer.

Overall, these analyses on price impact/realized spread at various short horizons are informative. They suggest that greater AT intensity leads to improved liquidity mainly through reducing the price impact of trades rather than changing net revenue for liquidity suppliers.

V. Additional Analysis

In this section, we conduct two additional tests to support our main analysis. First, we carry out market-level panel regression analysis among all 42 markets using co-location as the instrument variable. Second, we examine the relation between AT intensity and market quality in each market and draw inferences based on 42 estimates of coefficients.

A. Market-level IV Estimation

To complement our firm-level analysis, we also conduct market-level estimation with the adoption of co-location as an instrumental variable for AT. Specifically, each day, we aggregate all variables within each market by forming value-weighted averages across firms. This produces one time-series for each market. To maintain a balanced panel with all 42 markets, this analysis uses data from 2005 to 2011. We estimate a two-way panel across markets and days using co-location as the instrument for AT. In the first stage, we regress AT on the co-location indicator variable that equals one once that market officially starts co-location service. For markets that have not adopted co-location services by the end of our sample period, this indicator variable has a value of zero. As expected, co-location significantly contributes to AT activity, with a highly significant coefficient of 0.217 ($t\text{-stat} = 27.9$). In the second stage, we use the predicted value of AT as a regressor. While such market-level IV estimation allows us to conduct a difference in differences analysis, it has lower statistical power.

The second-stage results are reported in Table 8. Overall, these market-level estimates deliver results similar to the firm-level analysis. Importantly, despite the lower power of this approach, most estimates remain significant. The liquidity measures in Panel A decline as AT increases, suggesting that more AT leads to narrower spreads. The efficiency measures in Panel B also decrease with higher AT activity, likewise implying greater efficiency. Panel C shows that AT increases volatility. In Panel D, we observe better execution quality for buy-side investors who incur smaller execution shortfalls when AT increases.

B. Market-specific Analysis

We also use another approach to help draw inferences across markets. Specifically, we first conduct country-specific analysis on the relation between AT intensity and market quality for each of the 42 markets, and then aggregate the results across markets. Cross-market inference is based on equal-weighted means of the 42 market-specific AT coefficients and simple cross-sectional t-statistics also based on these 42 observations. Although this approach does not directly address causality, we can still learn from the actual (not instrumented) relationships between AT and market quality, especially when the relationships are consistent across countries. We find that better liquidity, improved price efficiency, and higher volatility are observed when AT is high. These results are reported in Appendix 2.

VI. Conclusion

Previous studies have produced mixed evidence on the effect of AT/HFT on market quality. This is partly due to data limitations that constrain researchers to either a small sample of stocks or a specific market outside the U.S. We examine a large international sample over an extended period of time that helps draw broader inferences about the effect of AT on market quality.

Surprisingly consistent across the 42 markets in our sample, more intense algorithmic trading leads to improved liquidity, better price efficiency, and elevated volatility. The overall net effect of AT on buy-side institutional investors results in lower execution shortfalls, suggesting that more AT benefits this group of market participants. We further show that AT's effect on volatility cannot be attributed to more efficient prices that adjust faster to new information, to the activities of liquidity suppliers seeking out more volatile stocks, or to heightened volatility at news announcements.

Aside from AT's overall influence on market quality, its effects are not uniform across all stocks. Large stocks enjoy more benefits from AT on liquidity and efficiency improvement, and small stocks see the highest AT-induced volatility. These results indicate some segmentation of liquidity provision across stocks of different sizes in our sample.

Overall, our findings support prior results that attribute liquidity-enhancing and efficiency-enhancing effects to AT/HFT activity. We show that this finding is surprisingly pervasive across countries. We complement prior studies with evidence that AT's liquidity provision does not apply uniformly to all firms, suggesting that regulation should tread carefully and recognize that the effects of AT differ across firm characteristics. Equally important, we show that more AT increases volatility, thereby imposing costs on market participants.

These results suggest that measures of the true cost of algorithmic trading should incorporate realized volatility. Our evidence suggests that the effect of AT on volatility is unlikely to derive from reverse causality and doesn't rely on specific types of volatility that market participants would consider beneficial. This suggests caution when it comes to regulating AT/HFT to produce a level playing field.

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FIGURE 1

Messages and AT

Figure 1 presents the time series of aggregate messages and AT measure. Our sample covers the period 2001-2011. Panel A graphs the time series of messages. We count all intraday messages that represent trades or changes in the price or size of the best quotes for each stock. Panel B plots the times series of AT. AT is the negative of dollar trading volume (\$100) per message. Then we compute equally weighted averages each market month, and report the mean across 42 markets.

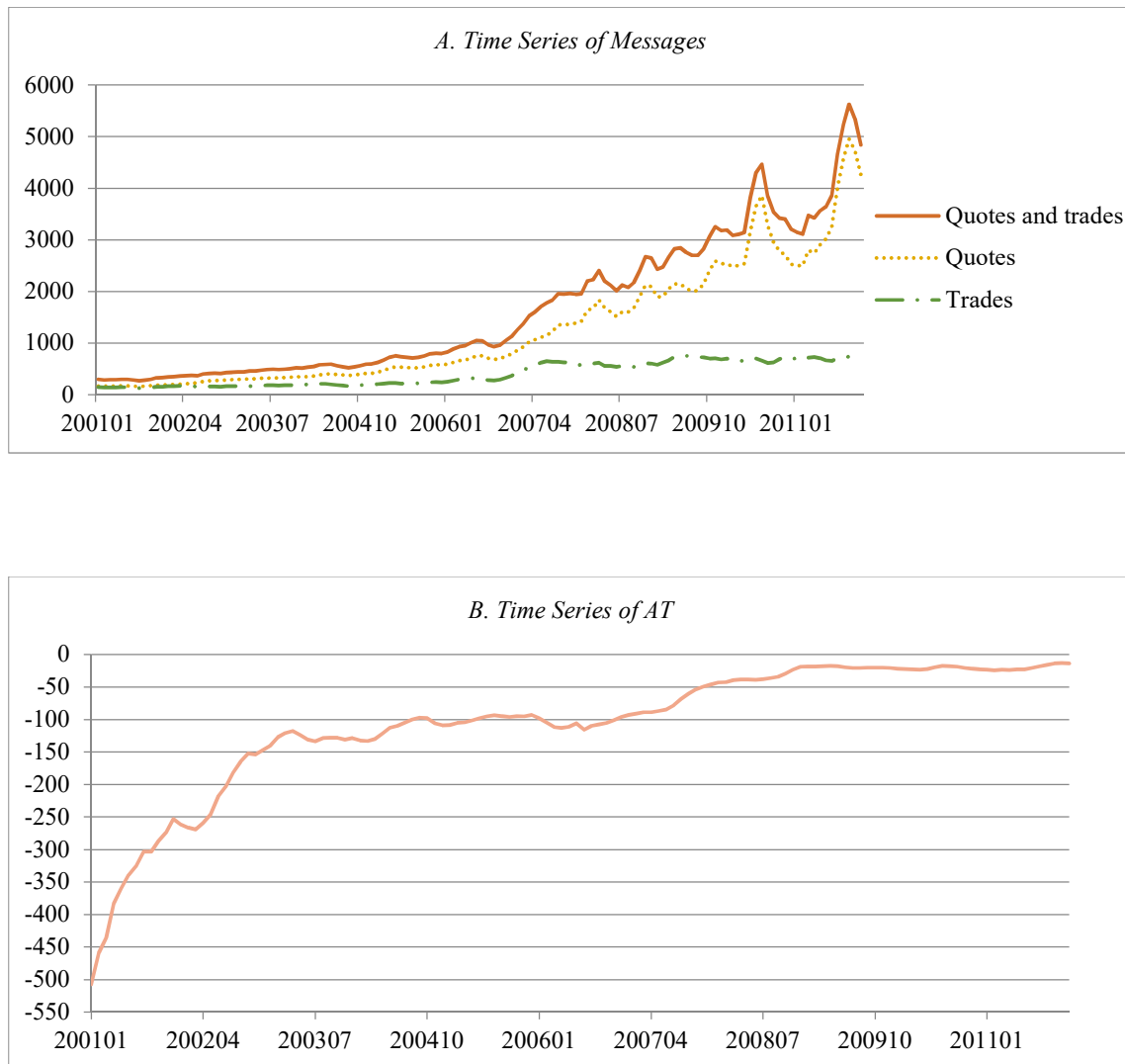


FIGURE 2

Market Quality Measures Over Time

Figure 2 graphs the time series of market quality over the sample period. All measures are computed intraday for each stock. Then we compute the mean for each day, and then the mean for each market month. The figures report the average across 42 markets. Our sample covers the period 2001-2011. Panel A plots time weighted relative quoted spreads (RQS) and relative effective spreads (RES) in basis points (bps). Panel B plots $|AR10|$ ($|AR30|$) defined as the absolute value of the daily average 10-minute (30-minute) quote-midpoint return autocorrelations. Panel C plots the daily intraday price range standardized by the daily closing price (PRICE_RANGE) and the variances of 30-minute quote midpoint returns (RET30_VAR). Panel D plots execution shortfall estimated following Anand et al. (2012).

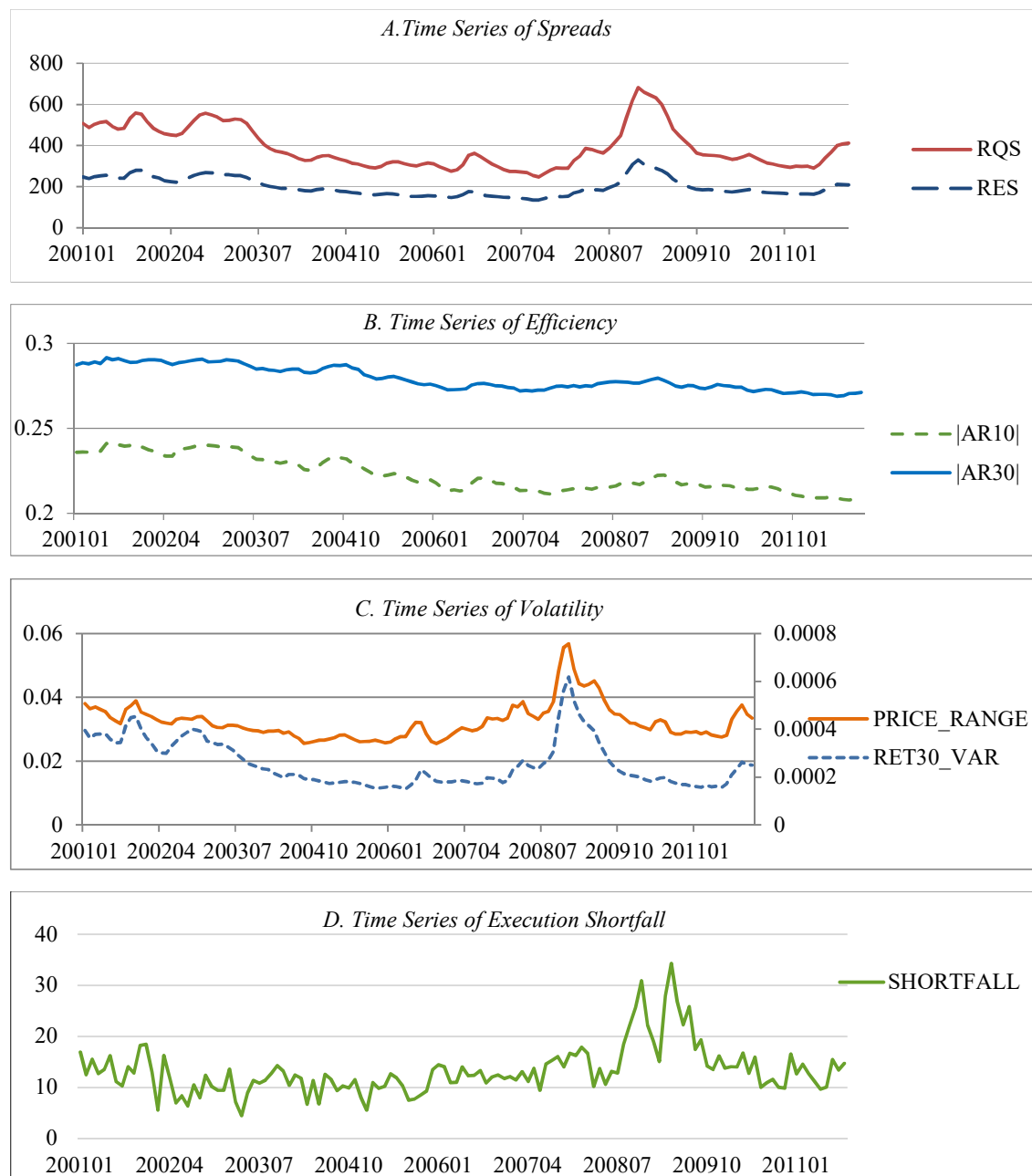


TABLE 1
Number of Stocks Listed

Table 1 lists the year-end number of stocks listed on each stock exchange in our sample. The sample period is 2001-2011. Growth is calculated as the number of firms in 2011 divided by the number of firms in the beginning of the sample and then minus 1.

Market	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	Growth
Argentina	19	55	64	45	61	66	68	57	65	71	65	2.42
Athens	287	285	287	284	276	263	250	226	219	186	168	-0.41
Australia	714	666	921	990	1105	1343	1415	1215	1337	1254	1220	0.71
Brussels	121	105	124	117	132	132	139	128	139	130	131	0.08
Copenhagen	154	143	164	169	166	183	207	201	202	193	177	0.15
Xetra	239	264	257	264	314	345	377	368	366	381	383	0.60
Euronext												
Amsterdam	152	148	139	136	136	138	129	116	114	108	104	-0.32
Euronext Lisbon					42	41	41	43	42	40	35	0.83
Euronext Paris	480	462	456	454	567	635	669	611	615	585	575	0.20
Helsinki	152	149	143	148	147	148	147	141	138	134	129	-0.15
Hong Kong	389	419	493	520	549	580	629	701	765	832	802	1.06
Istanbul	277	286	287	299	305	314	313	307	314	338	359	0.30
Jakarta	261	253	277	274	267	280	295	185	304	327	347	0.33
Johannesburg	270	220	230	224	245	263	299	266	257	256	261	-0.03
Korea	311	300	667	670	678	703	713	742	711	754	764	1.46
Kuala Lumpur	692	728	782	824	857	870	855	811	824	829	813	0.17
London	838	781	821	892	1006	1334	905	799	735	891	736	-0.12
Madrid	120	119	110	110	108	116	125	123	120	117	114	-0.05
Mexican	69	61	67	66	72	82	81	77	91	97	87	0.26
Milan	271	280	265	264	280	289	310	281	274	272	264	-0.03
Mumbai	240	425	716	821	1077	1161	999	939	1348	1414	1242	4.18
NASDAQ	3606	3237	2948	2879	2822	2797	2743	2622	2466	2359	2237	-0.38
NSE (India)	500	542	598	671	750	880	1034	1107	1188	1329	1308	1.62
New Zealand		55	54	63	77	82	95	74	84	86	69	1.25
NYSE	1524	1498	1475	1475	1459	1443	1392	1347	1328	1330	1311	-0.14
Osaka	183	195	232	242	277	275	263	250	253	252	246	0.34
Oslo	174	174	164	172	198	206	236	226	218	213	202	0.16
Philippines	140	112	135	143	153	184	187	162	195	194	205	0.46
Santiago		81	69	84	84	90	91	92	100	89	100	1.23
Sao Paulo						306	400	379	385	368	350	1.14
Shanghai	580	669	742	803	709	716	734	754	821	843	833	0.44
Shenzhen	459	466	487	516	459	508	596	693	780	1105	1314	1.86
Singapore	315	326	368	424	472	514	553	525	552	572	545	0.73
Stockholm	328	315	314	319	345	384	443	462	468	472	467	0.42
Swiss	224	218	228	227	235	235	227	227	221	218	213	-0.05
Taiwan	518	592	636	667	661	670	676	704	735	755	765	0.48
Tel-Aviv	310	283	319	342	417	469	528	495	484	480	453	0.46
Thailand	314	335	363	407	446	466	467	480	480	486	487	0.55
Tokyo	1957	1996	2072	2205	2310	2371	2360	2289	2284	2286	2261	0.16
Toronto	614	619	658	712	780	872	930	936	913	915	942	0.53
Warsaw	137	135	137	179	209	237	307	323	354	366	369	1.69
Wiener Borse	45	61	53	50	54	61	71	73	68	56	58	0.29

TABLE 2

Number of Messages per Stock-day by Market

Table 2 reports the summary statistics of messages at the beginning and end of our sample period. Messages represent trades or changes in the price or size of the best quotes in each stock. Reported numbers are the median of the daily mean of messages for each stock within each market. Growth is calculated as messages in 2011 divided by messages in 2001 and then minus 1. * Quote messages computed based on only price changes because quote sizes are not available until recent years. ** Data begin in 2005. *** Data begin in 2002. **** Data begin in 2006.

Market	Messages 2001	Messages 2011	Growth
Euronext Amsterdam	480	18718	38.00
Athens	324	245	-0.24
Australian	95	1403	13.82
Argentina	89	90	0.01
Thailand	219	551	1.52
Mumbai	120	132	0.10
Brussels	110	4375	38.63
Copenhagen	49	1039	20.33
Xetra	229	3597	14.68
Helsinki	233	2565	10.01
Hong Kong	247	1013	3.11
Istanbul	246	970	2.94
Johannesburg	74	2232	29.32
Jakarta	115	689	4.98
Kuala Lumpur	85	200	1.35
Korea	523	3547	5.79
London*	259	2171	7.38
Euronext Lisbon**	281	3167	10.25
Madrid	1014	8649	7.53
Milan	553	5621	9.16
Mexican	113	2600	21.91
NASDAQ	1006	15363	14.27
NSE (India)	357	2768	6.75
NYSE	1207	45998	37.10
New Zealand ***	33	66	0.97
Oslo	119	2002	15.85
Osaka	73	468	5.37
Euronext Paris	566	9266	15.37
Philippines	35	159	3.56
Swiss	99	688	5.92
Sao Paulo ****	621	4599	6.40
Singapore	170	279	0.64
Santiago ***	19	134	6.04
Shanghai	489	7710	14.78
Stockholm	242	1791	6.40
Shenzhen	431	2522	4.85
Tokyo	280	2026	6.24
Tel-aviv	55	549	8.90
Toronto	221	7659	33.70
Taiwan	487	656	0.35
Wiener Borse	60	1273	20.27
Warsaw	100	200	1.00

TABLE 3

The Effect of AT on Market Quality

Table 3 reports AT's effect on market quality using panel regression, with co-location as instrument variable for AT. In the 1st stage, AT is regressed on CO_LOCATION indicator variable. In the 2nd stage, each measure of market quality is regressed on predicted AT (AT*) and controls. Panel A presents the effect of AT on liquidity measures including time-weighted quoted spread (RQS) and trade-weighted relative effective spread (RES). Panel B presents the effect of AT on efficiency measures including the absolute value of intraday autocorrelations ($|AR_{\#}|$), measured for quote-midpoint returns over 10-minute and 30-minute periods. Panel C presents the effect of AT on volatility measures including daily price range standardized by the closing price (PRICE_RANGE), log value of the daily variances of 10-minute and 30-minute quote midpoint returns ($\ln(RET_{\#_VAR})$), and the absolute value of daily return ($|RET|$). Panel D presents the effect of AT on execution SHORTFALL calculated following Anand et al. (2012). AT is the negative of dollar trading volume (\$100) per message where messages include all inside quote changes and trade messages. Control variables include daily share turnover (TURN_OVER), PRICE_RANGE, the inverse of share price (INVERSE_PRC), the natural log of market capitalization ($\ln(MKT_CAP)$), and the first lag of the dependent variable, all measured at t-1. Regressions where the dependent variable is volatility do not include PRICE_RANGE, but add RES and $|AR30|$. The estimation is performed for the 12 months before and 12 months after the co-location in the 22 markets that adopted this service. All continuous variables are standardized to have a zero mean and unit variance within the estimation window for each firm. Standard errors are clustered by date. The table reports coefficients on AT*. *, **, *** indicate significance at 10%, 5% and 1%, respectively.

Panel A. Liquidity

	Coef.	t-stat
RQS	-0.220**	-2.35
RES	-0.705***	-13.93

Panel B. Efficiency

	Coef.	t-stat
$ AR10 $	-0.320***	-3.45
$ AR30 $	-0.227***	-2.45

Panel C. Volatility

	Coef.	t-stat
PRICE_RANGE	1.246***	4.73
$\ln(RET10_VAR)$	1.564***	6.01
$\ln(RET30_VAR)$	1.438***	5.61
$ RET $	0.385**	2.41

Panel D. Execution Shortfall

	Coef.	t-stat
SHORTFALL	-0.116**	-2.44

TABLE 4

The Effect of AT on Market Quality by Size

Table 4 reports how AT's effect on market quality varies with size, using a panel regression with co-location as instrument variable for AT. In the 1st stage, we regress AT on CO_LOCATION indicator variable, and in the 2nd stage, each measure of market quality is regressed on predicted AT (AT^*), its interaction with three size indicator variables S , M , and L , and controls. Panel A presents the effect of AT on liquidity measures including time-weighted quoted spread (RQS) and trade-weighted relative effective spread (RES). Panel B presents the effect of AT on efficiency measures including the absolute value of intraday autocorrelations ($|AR\#\#|$), measured for quote-midpoint returns over 10-minute and 30-minute periods. Panel C presents the effect of AT on volatility measures including daily price range standardized by the closing price (PRICE_RANGE), log value of the daily variances of 10-minute and 30-minute quote midpoint returns ($\ln(RET\#\#_VAR)$), and the absolute value of daily return ($|RET|$). Panel D presents the effect of AT on execution SHORTFALL calculated following Anand et al. (2012). AT is the negative of dollar trading volume (\$100) per message where messages include all inside quote changes and trade messages. Control variables include daily share turnover (TURN_OVER), PRICE_RANGE, the inverse of share price (INVERSE_PRC), the natural log of market capitalization ($\ln(MKT_CAP)$), and the first lag of the dependent variable, all measured at $t-1$. Regressions where the dependent variable is volatility do not include PRICE_RANGE, but add RES and $|AR30|$. The estimation is performed for the 12 months before and 12 months after the co-location in the 22 markets that already adopted co-location. All continuous variables are standardized to have a zero mean and unit variance within the estimation window for each firm. Standard errors are clustered by date. *, **, *** indicate significance at 10%, 5% and 1%, respectively.

Panel A. Liquidity

		Coef.	t-stat
RQS	$AT^* \times S$	-0.212***	-3.19
	$AT^* \times M$	-0.147	-1.61
	$AT^* \times L$	-0.288**	-2.36
RES	$AT^* \times S$	-0.479***	-10.02
	$AT^* \times M$	-0.532***	-10.25
	$AT^* \times L$	-0.998***	-15.47

Panel B. Efficiency

		Coef.	t-stat
$ AR10 $	$AT^* \times S$	-0.069	-0.65
	$AT^* \times M$	-0.497***	-4.79
	$AT^* \times L$	-0.322***	-2.76
$ AR30 $	$AT^* \times S$	0.054	0.52
	$AT^* \times M$	-0.324***	-3.22
	$AT^* \times L$	-0.301**	-2.37

Panel C. Volatility

		Coef.	t-stat
PRICE_RANGE	$AT^* \times S$	1.150***	5.03
	$AT^* \times M$	1.059***	4.35
	$AT^* \times L$	1.512***	4.84
$\ln(RET10_VAR)$	$AT^* \times S$	1.879***	8.35
	$AT^* \times M$	1.320***	5.34

ln(RET30_VAR)	$AT^* \times L$	1.499***	4.78
	$AT^* \times S$	1.642***	7.33
	$AT^* \times M$	1.287***	5.33
RET	$AT^* \times L$	1.330***	4.30
	$AT^* \times S$	0.692***	4.22
	$AT^* \times M$	0.150	0.88
	$AT^* \times L$	0.159	0.89

Panel D. Execution Shortfall

		Coef.	t-stat
SHORTFALL	$AT^* \times S$	-0.023	-0.13
	$AT^* \times M$	-0.132**	-2.12
	$AT^* \times L$	-0.126***	-3.04

TABLE 5

Correlation Between AT Coefficients in Liquidity and Volatility Regressions

In Table 5, we first estimate cross-sectional regressions each day with each market to obtain AT coefficients. We regress liquidity, efficiency, and volatility measures on AT and controls. Liquidity measures include time-weighted quoted spread (RQS), trade-weighted relative effective spread (RES). Efficiency measures are daily observations of the absolute value of intraday autocorrelations ($|AR_{\#}|$), measured for quote-midpoint returns over 10-minute and 30-minute periods. Volatility measures include daily price range standardized by the closing price (PRICE_RANGE), log value of the daily variances of 10-minute and 30-minute quote midpoint returns ($\ln(RET_{\#_VAR})$), and the absolute value of return ($|RET|$). AT is the negative of dollar trading volume (\$100) per message where messages include all inside quote changes and trade messages. Control variables include daily share turnover (TURN_OVER), PRICE_RANGE, the inverse of share price (INVERSE_PRC), the natural log of market capitalization ($\ln(MKT_CAP)$), and the first lag of the dependent variable, all measured at t-1. Regressions where the dependent variable is volatility do not include PRICE_RANGE, but add RES and $|AR_{30}|$. All continuous variables are standardized every day to have zero mean and unit variance within each exchange. This table reports the mean spearman rank correlation between AT coefficients from the regressions of volatility and liquidity (Panel A) and between the AT coefficients from the regressions of volatility and efficiency (Panel B). *, **, *** indicate significance at 10%, 5% and 1%, respectively.

Panel A. Correlation of AT Coefficients from the Regressions of Volatility and Liquidity

	PRICE_RANGE		$\ln(RET_{10_VAR})$		$\ln(RET_{30_VAR})$		$ RET $	
RQS	0.04	***	0.10	***	0.08	***	0.01	*
RES	0.08	***	0.14	***	0.11	***	0.06	***

Panel B. Correlation of AT Coefficients from the Regressions of Volatility and Efficiency

	PRICE_RANGE		$\ln(RET_{10_VAR})$		$\ln(RET_{30_VAR})$		$ RET $	
$ AR_{10} $	-0.10	***	0.04	**	-0.09	***	-0.04	***
$ AR_{30} $	-0.05	***	0.01		0.01		-0.02	***

TABLE 6

The Effect of AT on Volatility on News Days

Table 6 reports how AT's effect on volatility varies with news, using a panel regression with co-location as instrument variable for AT. In the 1st stage, we regress AT on CO_LOCATION indicator variable, and in the 2nd stage, each measure of volatility is regressed on predicted AT (AT^*), its interaction with news indicator (NEWS), and controls. Volatility measures include daily price range standardized by the closing price (PRICE_RANGE), log value of the daily variances of 10-minute and 30-minute quote midpoint returns ($\ln(R\#_VAR)$), and the absolute value of return ($|RET|$). AT is the negative of dollar trading volume (\$100) per message where messages include all inside quote changes and trade messages. Control variables include daily share turnover (TURN_OVER), the inverse of share price (INVERSE_PRC), the natural log of market capitalization ($\ln(MKT_CAP)$), and the first lag of the dependent variable, trade-weighted relative effective spread (RES), absolute value of intraday autocorrelations measured for quote-midpoint returns over 30-minute period ($|AR30|$), all measured at t-1. The estimation is performed for the 12 months before and 12 months after the co-location in the 22 markets that already adopted co-location. All continuous variables are standardized to have zero mean and unit variance within the estimation window for each firm. Standard errors are clustered by date. *, **, *** indicate significance at 10%, 5% and 1%, respectively.

		Coef.	t-stat
PRICE_RANGE	AT^*	1.171***	5.10
	$AT^* \times NEWS$	1.524**	2.29
	NEWS	0.079***	4.31
$\ln(R\#_VAR)$	AT^*	1.392***	5.91
	$AT^* \times NEWS$	2.292***	4.10
	NEWS	0.053***	3.56
$\ln(R\#_VAR)$	AT^*	1.262***	5.44
	$AT^* \times NEWS$	1.954***	3.87
	NEWS	0.044***	3.23
$ RET $	AT^*	0.291*	1.84
	$AT^* \times NEWS$	-0.354	-1.07
	NEWS	0.120***	13.53

TABLE 7

The Effect of AT on Relative Price Impact and Relative Realized Spread

Table 7 reports AT's effect on permanent price impact (RPI) and temporary price impact (RRS) using panel regression, with co-location as instrument variable for AT. In the first stage, AT is regressed on CO_LOCATION indicator variable. In the second stage, RPI (RRS) estimated at 30, 60, 90, 120 seconds following a trade is regressed on predicted AT and controls. Control variables include daily share turnover (TURN_OVER), daily price range standardized by the closing price (PRICE_RANGE), the inverse of share price (INVERSE_PRC), the natural log of market capitalization (ln(MKT_CAP)), and the first lag of the dependent variable, all measured at t-1. The estimation is performed for the 12 months before and 12 months after the co-location in the 22 markets that adopted this service. All continuous variables are standardized to have zero mean and unit variance within the estimation window for each firm. Standard errors are clustered by date. The table reports coefficients on predicted AT. *, **, *** indicate significance at 10%, 5% and 1%, respectively.

	Coef	t-stat
<i>Panel A. RPI as the Dependent Variable</i>		
RPI_30	-0.482***	-13.15
RPI_60	-0.476***	-13.19
RPI_90	-0.465***	-12.94
RPI_120	-0.457***	-12.70
<i>Panel B. RRS as the Dependent Variable</i>		
RRS_30	-0.065*	-1.81
RRS_60	-0.006	-0.16
RRS_90	0.027	0.83
RRS_120	0.047	1.48

TABLE 8

The Effect of AT on Market Quality at the Aggregate Level

Table 8 reports the effect of AT on market quality at the market level. Each day, we aggregate all variables within each market by forming market-value-weighted averages across firms. We estimate a two-way panel across markets and days using an instrumental variable approach. We use co-location as an instrument for algorithmic trading, a market-specific dummy that switches on once that market officially or at least publicly starts co-location service. Panel A examines liquidity measures including time-weighted quoted spread (RQS) and trade-weighted relative effective spread (RES). Panel B examines efficiency measures including the absolute value of intraday autocorrelations ($|AR_{\#}|$), measured for quote-midpoint returns over 10-minute and 30-minute periods. Panel C examines volatility measures including daily price range standardized by the closing price (PRICE_RANGE), log value of the daily variances of 10-minute and 30-minute quote midpoint returns ($\ln(RET_{\#_VAR})$), and the absolute value of return ($|RET|$). Panel D examines execution SHORTFALL calculated following Anand et al. (2012). AT is the negative of dollar trading volume (\$100) per message where messages include all inside quote changes and trade messages. Control variables include daily share turnover (TURN_OVER), PRICE_RANGE, the inverse of share price (INVERSE_PRC), the natural log of market capitalization ($\ln(MKT_CAP)$), and the first lag of the dependent variable, all measured at t-1. Regressions where the dependent variable is volatility do not include PRICE_RANGE, but add RES and $|AR30|$. Regressions where the dependent variable is volatility do not include price range, but add RES and $|AR30|$. The sample period is from 2005 to 2011 to maintain a balanced panel where all 42 markets are present in the data. *, **, *** indicate significance at 10%, 5% and 1%, respectively.

Panel A. Liquidity

	Coef.	t-stat
RQS	-0.023***	-4.06
RES	-0.045***	-7.12

Panel B. Efficiency

	Coef.	t-stat
$ AR10 $	-0.041***	-4.00
$ AR30 $	0.010	0.99

Panel C. Volatility

	Coef.	t-stat
PRICE_RANGE	0.060***	9.99
$\ln(RET10_VAR)$	0.076***	15.62
$\ln(RET30_VAR)$	0.093***	16.79
$ RET $	0.036***	5.11

Panel D. Execution Shortfall

	Coef.	t-stat
SHORTFALL	-0.024**	-1.95

APPENDIX 1

Co-location Events in Our Sample

Appendix 1 lists the exchanges that adopted the co-location service in our sample period 2001-2011.

Country	Earliest co-location date	Link (as of September 2013)
Australia	200811	http://www.asxgroup.com.au/media/PDFs/mr030708_co-location_hosting.pdf
Belgium	200804	https://europeanequities.nyx.com/sites/europeanequities.nyx.com/files/327777.pdf
Brazil	20090629	http://ir.bmfbovespa.com.br/enu/1190/NMCoLocation.pdf
Canada	200811	http://www.investorpoint.com/stock/X%253ACA-TMX+Group+Limited/news/6495414
Denmark	20080625	http://www.ft.com/intl/cms/s/0/b2cde4f0-42ce-11dd-81d0-0000779fd2ac.html#axzz2RRv89HWs
Finland	20080625	http://www.ft.com/intl/cms/s/0/b2cde4f0-42ce-11dd-81d0-0000779fd2ac.html#axzz2RRv89HWs
France	200804	https://europeanequities.nyx.com/sites/europeanequities.nyx.com/files/327777.pdf
Germany	2006Q4	http://deutsche-boerse.com/dbg/dispatch/en/listcontent/gdb_navigation/press/10_Latest_Press_Releases/Content_Files/13_press/August_2006/pm_news_Proximity_090806.htm?newstitle=deutscheboerseysystemsandixeuro&location=press
India	20101115	http://www.business-standard.com/article/markets/bse-s-co-location-facility-begins-to-get-brokers-111012000030_1.html
India	200908	http://www.nseindia.com/circulars/circular.htm
Italy	200909	http://www.borsaitaliana.it/borsaitaliana/ufficio-stampa/comunicati-stampa/2008/081110avviotradelect.en_pdf.htm
Japan	200811	http://www.ose.or.jp/e/news/14734
Japan	200905	http://www.tse.or.jp/english/rules/equities/arrowhead/info.html
Netherlands	200804	https://europeanequities.nyx.com/sites/europeanequities.nyx.com/files/327777.pdf
Portugal	200804	https://europeanequities.nyx.com/sites/europeanequities.nyx.com/files/327777.pdf
Singapore	201104	http://www.sgx.com/wps/wcm/connect/sgx_en/home/highlights/news_releases/news+update+sgxs+reach+trading+engine+goes+live
Sweden	20080625	http://www.ft.com/intl/cms/s/0/b2cde4f0-42ce-11dd-81d0-0000779fd2ac.html#axzz2RRv89HWs
Switzerland	20080624	http://www.six-swiss-exchange.com/swx_messages/online/swx_message_200901161725_en.pdf
Taiwan	2010Q4	http://asiaetrading.com/taiwan-stock-exchange-to-launch-co-location-services/
UK	200809	http://www.londonstockexchange.com/about-the-exchange/media-relations/press-releases/2008/launchofexchangehostingcreatessub-millisecondaccessstoitsmarkets.htm
USA	200504	http://www.wallstreetandtech.com/electronic-trading/data-latency-playing-an-ever-increasing/199702208
USA	200701	http://www.watertechnology.com/inside-market-data/news/1614171/essex-radez-co-lo-cuts-ny-latency

APPENDIX 2

Relationship Between AT and Market Quality: Market-specific Analysis

Appendix 2 presents market-specific analysis. Our data cover 42 markets from 2001-2011. We first estimate, for each market, a firm and day fixed effects panel regression. We show the mean coefficients across the 42 markets, the associated t-statistic using the cross-market standard deviation, the percentage of positive and significant market-specific coefficients. Panel A examines liquidity measures including time-weighted quoted spread (RQS) and trade-weighted relative effective spread (RES). Panel B examines efficiency measures including the absolute value of intraday autocorrelations $|AR_{\#}|$, measured for quote-midpoint returns over 10-minute and 30-minute periods. Panel C examines volatility measures including daily price range standardized by the closing price (PRICE_RANGE)log value of the daily variances of 10-minute and 30-minute quote midpoint returns ($\ln(RET_{\#_VAR})$), and the absolute value of return ($|RET|$). Panel D examines execution SHORTFALL calculated following Anand et al. (2012). AT is the negative of dollar trading volume (\$100) per message where messages include all inside quote changes and trade messages. Control variables include daily share turnover (TURN_OVER), PRICE_RANGE, the inverse of share price (INVERSE_PRC), the natural log of market capitalization ($\ln(MKT_CAP)$), and the first lag of the dependent variable, all measured at t-1. Regressions where the dependent variable is volatility do not include PRICE_RANGE, but add RES and $|AR30|$. All continuous variables are standardized every day to have a zero mean and unit variance within each exchange.

Panel A. Liquidity

	Coef.	t-stat	Percent positive and significant
RQS	-0.009***	-6.69	2%
RES	-0.010***	-3.52	26%

Panel B. Efficiency

	Coef.	t-stat	Percent positive and significant
$ AR10 $	-0.014***	-7.23	7%
$ AR30 $	-0.004***	-4.01	14%

Panel C. Volatility

	Coef.	t-stat	Percent positive and significant
PRICE_RANGE	0.040***	8.52	83%
$\ln(RET10_VAR)$	0.022***	4.25	71%
$\ln(RET30_VAR)$	0.030***	5.17	79%
$ RET $	0.027***	7.65	81%

Panel D. Execution Shortfall

	Coef.	t-stat	Percent positive and significant
SHORTFALL	-0.002	-0.58	3%